CH3 Conditional Prob

3.1 Conditional Probability

If P(B)>0, the conditional probability of A given B denotes: $P(A|B)=\frac{P(AB)}{P(B)}$

在 B 情況下，再發生 A 的機率

Ex 3.1 In a certain region of Russia,

P(one lives at least 80)=0.75, P(one lives at least 90)=0.63.

What is the probability that a randomly selected 80-year-old person from this will survive to become 90?

已經發生的事情：Lives at least 80 → Event B

要求計算的事情：Lives at least 90 → Event A $\quad P(A|B)=\frac{P(AB)}{P(B)}=\frac{0.63}{0.75}$

Ex 3.5 Draw 8 cards from a deck of 52 cards.

Given 3 of them are spades, P(the remaining 5 are also spades)=?

B: the event that at least 3 of them are spades
A: the event that all 8 cards selected are spades
P(A|B)=P(AB)/P(B)=P(A)/P(B)
P(A)=C(13,8)/C(52,8)
P(B)=C(13,3)C(39,5)/C(52,8)
+C(13,4)C(39,4)/C(52,8)+ …
+C(13,8)C(39,0)/C(52,8)
P(A|B)=5.44 x 10^{-6}

Theorem 3.1

Let S be the sample space of an experiment, and let B be an event of S with P(B)>0. Then

(a) $P(A\mid B)\geq 0$ for any event A of S

(b) $P(S\mid B)=1$

(c) If $A_1, A_2, \ldots$ is a sequence of mutually exclusive events,

then $P(\bigcup_{i=1}^{\infty} A_i\mid B)=\sum_{i=1}^{\infty} P(A_i\mid B)$

A Few useful Formula

1. $P(A\mid B)=1-P(A^c\mid B)$

2. $P(A\cup C\mid B)=P(A\mid B)+P(C\mid B)-P(AC\mid B)$

3. $P(A\mid B)=P(AC\mid B)+P(AC^c\mid B)$

3.2 Law of multiplication

1. $P(A \mid B) = \frac{P(AB)}{P(B)}$
2. $\Rightarrow P(AB) = P(A \mid B)P(B)$
3. $P(AB) = P(BA) = P(B \mid A)P(A)$

Ex 3.9 Suppose 5 good fuses and 2 defective ones have been mixed up. To find the defective fuses, test them 1-by-1, at random and without replacement. What is the probability that we are lucky and find both of the defective fuses in the first two tests?

D_1 : The first time we'v got a dead one$= \frac{2}{7}$

D_2 : The second time we'v got a dead one$= \frac{1}{6}$

$P(D_1 D_2) = P(D_1) \times P(D_1|D_2) = \frac{2}{7} \times \frac{1}{6} = \frac{1}{21}$

Ex 3.11 Suppose 5 good fuses and 2 defective ones have been mixed up. To find the defective fuses, we test them 1-by-1, at random and without replacemen What is the probability that we find both of the defective fuses in exactl three tests?

Event 1: Good Bad Bad $= \frac{5}{7} \times \frac{2}{6} \times \frac{1}{5}$

Event 2: Bad Good Bad $= \frac{2}{7} \times \frac{5}{6} \times \frac{1}{5}$

Ans: $P(\text{Event 1} \cup \text{Event 2}) = P(E_1) + P(E_2) = \ldots$

3.3 Law of total probability

Theorem 3.3(Law of total probability)

$$P(A) = P(AB) + P(AB^c)$$
$$= P(A|B)P(B) + P(A|B^c)P(B^c)$$

Ex 3.12 An insurance company rents 35% of the cars for its customers from agency I and 65% from agency II. If 8% of the cars of agency I and 5% of the cars of agency II break down during the rental periods, what is the probability that a car rented by this insurance company breaks down?

Let A be a event that car rent from agency I

Let B be a event that a car break

$P(A) = 35\%$ $\quad P(B|A) = 8\%$

$P(A^c) = 65\%$ $\quad P(B|A^c) = 5\%$

$$P(B) = P(BA) + P(BA^c)$$
$$= P(B|A)\cdot P(A) + P(B|A^c)\cdot P(A^c) \cdots$$

Ex 3.14 (Gambler's ruin problem)

2 gamblers play the game of "heads or tails," in which each time a fair coin lands heads up, player A wins $1 from B, and each time it lands tails up, player B wins $1 from A. Suppose that the player A initially has a dollars and player B has b dollars. If they continue to play this game successively, what is the probability that

(a) A will be ruined;

令 P_i 為 A 在 i 元下破產之機率

1. $P_0 = 1$, $P_{a+b} = 1$

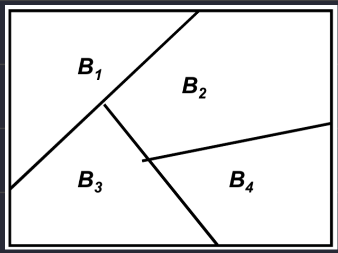
2. $P_i = \frac{1}{2} P_{i+1} + \frac{1}{2} P_{i-1}$

$P_{i+1} - P_i = P_i - P_{i-1}$

WIP

Definition: Partition

Let $\{B_1, B_2, B_3, ..., B_n\}$ be a nonempty subset of sample space S



1. All event B are mutually exclusive
2. $\cup_{i=1}^{n} B_i = S$

Then $\{B_1, B_2, B_3, ..., B_n\}$ is called the partition of S

Thm 3.3(Law of total probability)

If $\{B_1, B_2, B_3, ..., B_n\}$ be a partition of S, then

$$P(A) = P(AB_1) + P(AB_2) + P(AB_3) + ... + P(AB_n)$$

$$= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) + ... + P(A|B_n)P(B_n)$$

Ex 3.15 Suppose that 80% of the seniors, 70% of the juniors, 50% of the sophomores and 30% of the freshmen of a college use the library of their campus frequently. If 30% of all students are freshmen, 25% are sophomores, 25% are juniors, and 20% are seniors, what percent of all students use the library frequently?

Let B_1, B_2, B_3, B_4 repsent student of freshmen, sophomores, juniors, seniors

Let A be a student using library

$P(A) = P(A|B_1) + P(A|B_2) + P(A|B_3) + P(A|B_4)$

$= P(AB_1) \cdot P(B_1) + P(AB_2) \cdot P(B_2) + \cdots + P(AB_4) \cdot P(B_4)$

Ex 3.17 A box contains 10 white and 12 red balls. Two balls are drawn at random and, without looking at their colors, are discarded. What is the probability that the 3rd ball withdrawn is red?

$B_1 =$ ○○ $B_2 =$ ○○/○○ $B_3 =$ ○○ $A =$ ○○○

$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3)$

3.4 Bayes' formula (逆機率)

$$P(B_k \mid A) = \frac{P(AB_k)}{P(A)} = \frac{P(A \mid B_k)P(B_k)}{\sum_{i=1}^{n} P(AB_i)} = \frac{P(A \mid B_k)P(B_k)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}$$

$P(B_k)$ is called the prior probability of B_k (事前機率)

$P(B_k|A)$ is called the posterior probability of B_k after the occurrence of event A

Ex 3.21 A box contains 7 red and 13 blue balls. 2 balls are selected at random and are discarded without their colors being seen. If a 3 rd ball is drawn randomly and observed to be red, what is the probability that both of the discarded balls were blue?

Event A : The 3rd ball is red

Event B : 2 balls are selected were blue

$P(BB) = \frac{13}{20} \times \frac{12}{19}$

$P(RR) = \frac{7}{20} \times \frac{6}{19}$

$P(BR) = \frac{13}{20} \times \frac{7}{19} + \frac{7}{20} \times \frac{13}{19}$

$$P(B|A) = \frac{P(AB)}{P(A)} = \frac{P(A|B) \cdot P(B)}{P(A)}$$

$$= \frac{P(A|B) \cdot P(B)}{P(A|BB) \cdot P(BB) + P(A|BR) \cdot P(BR) + P(A|RR)\ P(RR)}$$

Ex cont A lab blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a false positiv result for 1 percent of the healthy persons tested. If 0.5 percent of the population actually has the disease. What is the probability that a person has the disease given that the test result is positive?

Event A : Person has disease

Event B : The result is positive

$P(A) = 0.5\%$ $P(A^c) = 99.5\%$

$P(B|A^c) = 1\%$ $P(B|A) = 95\%$

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|A^c) \cdot P(A^c)}$$

Ex cont We have 3 cards identical in form except that both sides of the 1st card are red, both sides of the 2nd card are black, one side of the 3rd card is red and the other side black. Suppose one card is randomly selected and put down on the ground. If the side of the chosen card is red, what is the probability that the other side is black?

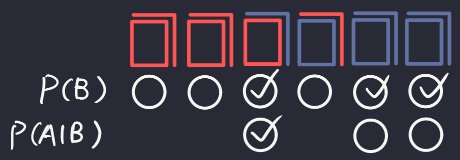
Event A : Side of the choosen card is red

Event B : Another side is black

$P(A) = \frac{3}{6} = \frac{1}{2}$

$P(A|B) \cdot P(B) = \frac{1}{3} \times \frac{3}{6}$

$$P(B|A) = \frac{P(BA)}{P(A)} = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$



3.5 Independence

If the conditional probability of A given B is equal to probability of A that is $P(A|B) = P(A)$, we say that A is independent of B

Ex 3.24 An urn contains 5 red and 7 blue balls. Suppose that 2 balls are selected at random and with(without) replacement. Let A and B be the events that the first and the second ball are red, respectively. Are A and B independent?

With : Independent

Without : Dependent

Ex 3.3 From the set of all families with 2 children, a child is selected at random and is found to be a girl. What is the probability that the second child of the girl's family is also a girl? Assume that in a 2-child family all sex distributions are equally probable.

$\frac{1}{2}$

Theorem 3.6 If A and B are independent, then A and B^c are independent as well.

Definition The events A, B, and C are called independent if

$P(BC) = P(B)P(C)$

$P(AB) = P(A)P(B)$

$P(AC) = P(A)P(C)$

☆ $P(ABC) = P(A)P(B)P(C)$

Definition: Bernoulli Trial

Consider a particular event A. The outcome of the Bernoulli trial is said to be a success if A occurs and a failure otherwise. We are interested in finding the probability of k successes in n independent repetitions of a Bernoulli trial.

→ 每次的機率都是獨立的，不會影響下次的 p

Sequence of Independent Events

1. Set the event E to the problem asking for
2. Split conditions into partitions that the problem provides
3. Calculate the union of E given by each partition
4. In the "Neither of the condition" situation, recursive might appear

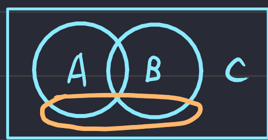
Ex 3.31 We draw cards, one at a time, at random and successively from an ordinary deck of 52 cards with replacement. What is the probability that an ace appears before a face card?

Event A: Got ace card first

Event B: Got face card first

Event C: Got neither of the above

Event E: Got ace before face



$$P(E) = P(E|A)\cdot P(A) + P(E|B)\cdot P(B) + P(E|C)\cdot P(C)$$

$$= 1\cdot\frac{4}{52} + 0\cdot\frac{12}{52} + P(E)\cdot\frac{36}{52}$$

$$P(E) = \frac{1}{4}$$

★ 非回合制

Ex 3.32 Consider a sequence of experiments where each experiment consists of two sub-experiments as follows. A and B alternates rolling a pair of dice, stopping either when A rolls the sum 9 or when B rolls the sum 6. Assume that A rolls first, find the probability that the final roll is made by A?

Event A: A rolls 9 first

Event B: A didn't roll 9, then B rolls 6

Event C: Neither of the above

Event E: A rolls 9 first while take turn

$$P(E) = P(E|A)\cdot P(A) + P(E|B)\cdot P(B) + P(E|C)\cdot P(C)$$

$$= 1\cdot\frac{4}{36} + 0\cdot\left(1-\frac{4}{36}\right)\cdot\frac{5}{36} + P(E)\cdot\left(1-P(A)-P(B)\right)$$

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